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Crystal structure and electrical properties of (Li, Ce, Nd)-multidoped $CaBi_2Nb_2O_9$ high temperature ceramics

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ABSTRACT

(Li, Ce, and Nd)-multidoped $CaBi_2Nb_2O_9$ (CBN) Aurivillius phase ceramics were prepared via a conventional solid-state sintering route. The crystal structure including bond lengths and bond angles, microstructure, dielectric constant, DC resistivity, and piezoelectric properties were systematically investigated. Rietveld-refinements of X-ray results indicated that small quantity of (Li, Ce, Nd) doping (< 2.5 mol%) increases orthorhombic distortion, because of the smaller ionic radii of doping ions. However, orthorhombic distortion obviously decreased with increasing (Li, Ce, Nd) doping concentration from 5 to 25 mol%. The replacement of asymmetric A-site Bi^{3+} with 6s2 lone pair electrons by symmetric Li^+, Ce^{3+} and Nd^{3+} ions decreased the orthorhombic distortion. The morphologies and electrical properties of sintered ceramics were tailored by the introducing (Li, Ce, Nd) multi-dopants. The improvement of piezoelectric properties of modified-CBN ceramics were attributed to decreasing grain sizes and morphotropic phase boundary (MPB). $Ca_{0.85}(Li_{0.5}Ce_{0.25}Nd_{0.25})_{0.15}Bi_2Nb_2O_9$ (CBNLCN-15) ceramics had optimum properties, and d_{33} and T_c values were found to be ~ 13.1 pC/N and ~ 900 °C, respectively.

1. Introduction

In recent years, the aerospace, aircraft, and nuclear power industries have a desperate need for sensors, which can operate at 500 °C or higher temperature with a long mean time between failures, because sensors need to be as close as possible to engine component of interest for adequate sensitivity to monitor engine health and enable safer operation [1,2]. Aurivillius phase compounds, called as bismuth layered structural ferroelectrics (BLSFs), are a large family of important and interesting ferroelectrics, possessing some unique properties such as high Curie temperature and fatigue-free properties. These shining properties make Aurivillius compounds as promising candidates for high-temperature piezoelectric sensors operating in some hazardous environments [1–4]. Therefore, these compounds have attracted increasing attention in recent years. The structure of these compounds can be described by a general formula:

$(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$,

where A is mono-, di-, trivalent ion or a mixture, allowing dodecahedral coordination, B is a combination of transition-metal cations well suited for octahedral coordination, and m is the number of octahedral layers in the perovskite slab in the range of 1–6 [5].

Calcium bismuth niobate, $CaBi_2Nb_2O_9$ (CBN), is a typical Aurivillius-type material with $m = 2$ and possesses an extremely high Curie temperature of ~930 °C and is a promising candidate for applications in the high-temperature sensors [6]. However, piezoelectric constant of CBN is relatively poor, with a value of ~ 5 pC/N, because the orientation of its spontaneous polarization is restricted to *a-b* plane, hindering their practical applications. Much work has been reported to improve the piezoelectric properties of CBN, such as chemical tailoring [7,8], grain size engineering [9,10] and textured-ceramic preparation [11,12]. Chemical modification is an effective way to tailor electrical properties of piezoelectric ceramics. A-site substitution is an effective way to improve the piezoelectric properties of CBN, especially the co-substitution with alkali metal cations and rare-earth cations. With (Li, Ce) [8] and (K, La) [13] A-site co-substitution, the d_{33} values of $Ca_{0.88}(LiCe)_{0.04}Bi_2Nb_2O_9$ and $Ca_{0.9}(KLa)_{0.05}Bi_2Nb_2O_9$ ceramics are 13.3 and 15.8 pC/N, respectively. Moreover, two or more types of rare-earth elements modified CBN ceramics are rarely reported. Although Peng et al. [14] reported (Li, Ce, Pr) multi-doped $CaBi_2Nb_2O_9$ ceramics possessing a large d_{33} value (~ 17.3 pC/N), together with a high Curie temperature (T_C > 900 °C), the crystal structure of the modified CBN has not been reported. Therefore, studying the correlation between crystal structure and electric properties of two or more types of rare-earth elements doped CBN ceramics is significantly interesting.

In this study, (Li, Ce, Nd) multi-doped CBN ceramics were prepared

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via the conventional solid-state sintering method. Crystal structure, as well as detailed dielectric, piezoelectric, and electrical conduction properties was studied.

2. Experimental

Aurivillius type $Ca_{1-x}(Li_{0.5}Ce_{0.25}Nd_{0.25})_xBi_2Nb_2O_9$ (CBNLBN-100*x*, where x = 0, 0.025, 0.05, 0.1, 0.15, 0.2 and 0.25) ceramics were prepared by a conventional solid-state reaction route. Metal oxides, Bi_2O_3 (99%), $CaCO_3$ (99%), Nb_2O_5 (99.5%), Li_2CO_3 (98%), CeO_2 (99.9%), and Nd_2O_3 (99.9%) were used as raw materials. All powders were weighed according to stoichiometric compositions and then mixed by ball milling in ethanol for 6 h with ZrO_2 balls as milling media. The slurry was dried at 70 °C to remove ethanol and then calcined at 850 °C for 2 h. Calcined powders were milled again under the same conditions and granulated with polyvinyl alcohol (PVA) as a binder. The powders were finally pressed into disks of ~ 10 mm in diameter and ~ 1 mm thickness in a steel die at a pressure of ~ 5 T_f. After burning out the binder at 550 °C, the green pellets were sintered at 1050 °C for 2 h.

The crystal structure of the sintered ceramics was determined using an X-ray diffractometer (X'Pert PRO, PANalytical, Holland) equipped with Cu-Kα radiation (λ = 1.5418 Å). The microstructure of the sintered ceramics was characterized by scanning electron microscopy (SEM, S4800, Hitachi, Japan). Both the surfaces of the sintered ceramics were ground and polished into ~ 0.5-mm thick disk, and then fired Ag/Pd paste at 850 °C for 10 min as the electrodes. The dielectric behavior as a function of temperature was investigated using an LCR analyzer (IM3536, Hioki, Japan) attached to a programmable furnace. The temperature dependence of the electrical conductivity was measured over the temperature range from 300 to 700 °C at a heating rate of 5 °C/min. The specimens were poled in a silicone oil bath at 180 °C under a DC electric field of 5–12 kV/mm for 5–15 min. The piezoelectric coefficient d_{33} was measured using a quasi-static d_{33} meter (ZJ-3A, Institute of Acoustics, Chinese Academy of Sciences, China).

3. Results and discussion

Fig. 1 shows the XRD patterns of the CBN-based ceramics with various x contents. All the detected peaks are indexed and marked according to the JCPDS#49-0608. No secondary phase diffraction peaks are found in the range of measured 2θ angels, indicating that the (Li, Ce, Nd) dopants have entered into the CBN crystal cells to form the solid solution. All the diffraction peaks show the typical characteristic of two-layered Aurivillius phase compounds. The strongest diffraction peaks of the ceramics are the (115) plane in all the XRD patterns and are consistent with the fact that the most intense reflection of BLSFs is of the type (112m + 1) [15]. Furthermore, the enlarged XRD patterns in the 2θ ranges 29–30° and 32–34° are shown in Fig. 1(b) and (c). Notably, the diffraction peaks of the (200)/(020) plane merge into one peak, demonstrating a gradual change in the crystal structure from orthorhombic system to pseudo-tetragonal system with increasing (Li, Ce, Nd) doping concentration. There would be a morphotropic phase boundary (MPB) of orthorhombic and tetragonal phase near by 100x = 15. The slight shift in the diffraction peaks implies the variation in the lattice parameters, because of the introduction of (Li, Ce, Nd) dopants.

To further analyze the effect of (Li, Ce, Nd) doping on the crystal structure of CBNLCN-100*x* ceramics, XRD patterns of samples were analyzed by Rietveld refinement with the $A2_1am$ space group at room temperature, by using Fullprof software [16]. The crystal structure of CBN was reported previously by Blake et al. [17], where Bi^{3+} on the A site and Ca^{2+} on the Bi site of the $(Bi_2O_2)^{2+}$ layer were suggested. The coordinates of CBN reported by Blake et al. were selected as an initial model. Long et al. [18–20] reported a number of articles on A site doping, where A site substitutions occupy A and Bi sites of the $(Bi_2O_2)^{2+}$ layer. In this study, a small amount of (Li, Ce, Nd) doping only occupied A site; however, the observed and calculated intensities fitted well.

The Rietveld refined XRD patterns of CBNLCN-100*x* ceramics are shown in Fig. 2. The minimization is carried out using the reliability index parameters such as the residuals for the pattern R_p, the weighted pattern R_{wp}, and goodness of fit *Chi*2.

Fig. 3(a) shows the lattice parameter a, b, and c value of CBNLCN-100*x* ceramic calculated by the Rietveld refinement method. The orthorhombic structure stress (a/b) of BLSF can be used to evaluate the degree of the lattice distortion of pseudo-perovskite including BO_6 oxygen octahedral rotation and tilting [21]. With 100x increasing from 0 to 2.5, the lattice parameter a increases and b also slightly increases. Correspondingly, the value of a/b increases, as shown in Fig. 3(b), indicating that the orthorhombic structure of the CBN-type ceramics is enhanced. With 100x further increasing from 2.5 to 25, the value of a/b ratio decreases, and the crystal phase approaches tetragonal structure. The same phenomenon has been reported by Chen et al. [22].

The detailed atomic coordinates, atomic positions, and atomic occupancies of CBNLCN-100*x* ceramics are listed in Tables 1–6. Selected bond angles and bond lengths for CBNLCN-100*x* ceramics are listed in Table 7. According to the refinement results, the visualized crystal structures of CBNLCN-100*x* ceramics are shown in Fig. 4.

For Nb–O1–Nb bond angle in the NbO_6 octahedron, the angle decreases from 158.47° to 154.19° for CBN and CBNLCN-5. In addition, the difference between A–O1 bond lengths along a axis becomes more obvious (3.191 Å and 2.312 Å for CBN, 3.396 Å and 2.086 Å for CBNLCN-5), indicating that the tilt of the NbO_6 octahedron about the a axis increases. In contrast, with 100x further increasing from 5 to 25, the Nb–O1–Nb bond angle increases from 154.19° to 166.62°, and the

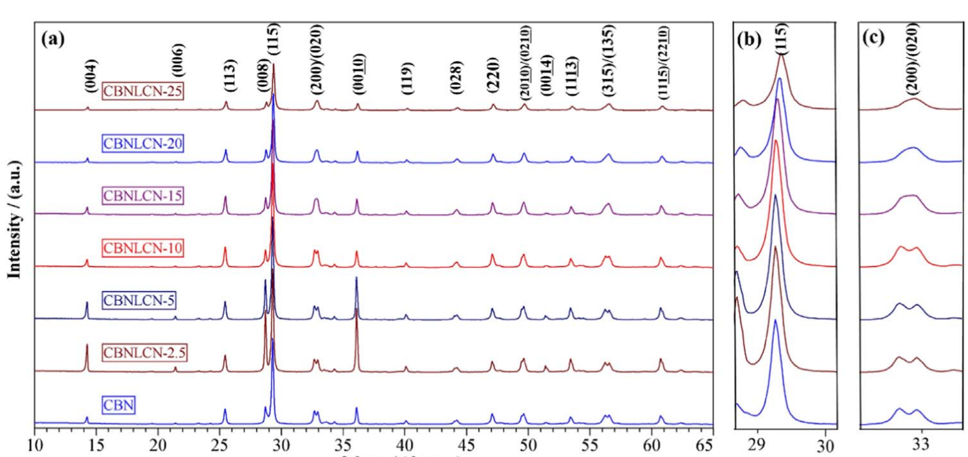

Fig. 1. (a) XRD patterns and (b) and (c) expanded XRD patterns of CBNLCN-100*x* ceramics.

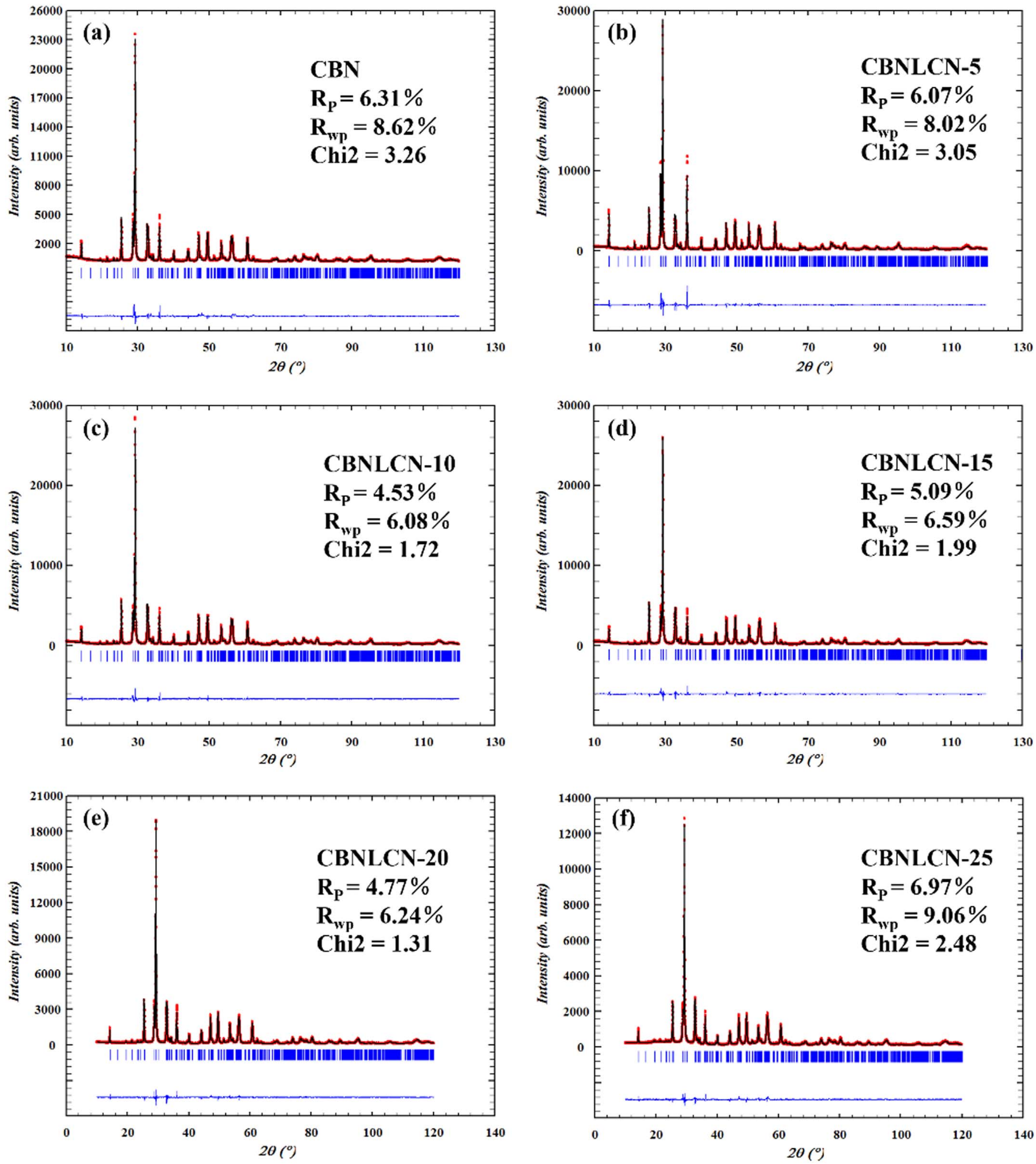

Fig. 2. Calculated (solid line), observed (dots) and difference (bottom) powder XRD patterns of CBNLCN-100x (x = 0, 0.05, 0.1, 0.15, 0.2 and 0.25) after structure refinements in the space group $A2_1am$. The vertical bars are the Bragg positions.

difference between the A–O1 bond lengths along a axis decreases, indicating that the tilt of the NbO_6 octahedron about the a axis decreases. In Aurivillius ferroelectrics, the structural distortion can be mainly described as the tilt of the NbO_6 octahedron about the a axis and the rotation of the NbO_6 octahedron along the c axis, where the neighboring octahedrons rotate in opposite directions [23]. The same change of bond angles has been observed in A-site substitution of CBN with smaller size Li^+, Ce^{3+}, and Nd^{3+} (1.15, 1.34, 1.27, and 1.34 Å for Li^+, Ce^{3+}, Nd^{3+}, and Ca^{2+}, respectively) [18,24]. A general trend is that the distortion of the structure increases with decreasing size of the A cation, and the insertion of a voluminous cation in the site tends to regularize it and therefore minimizes the tilt of the octahedron [25]. However, as the x value increases from 0.05 to 0.25, the occupancy of

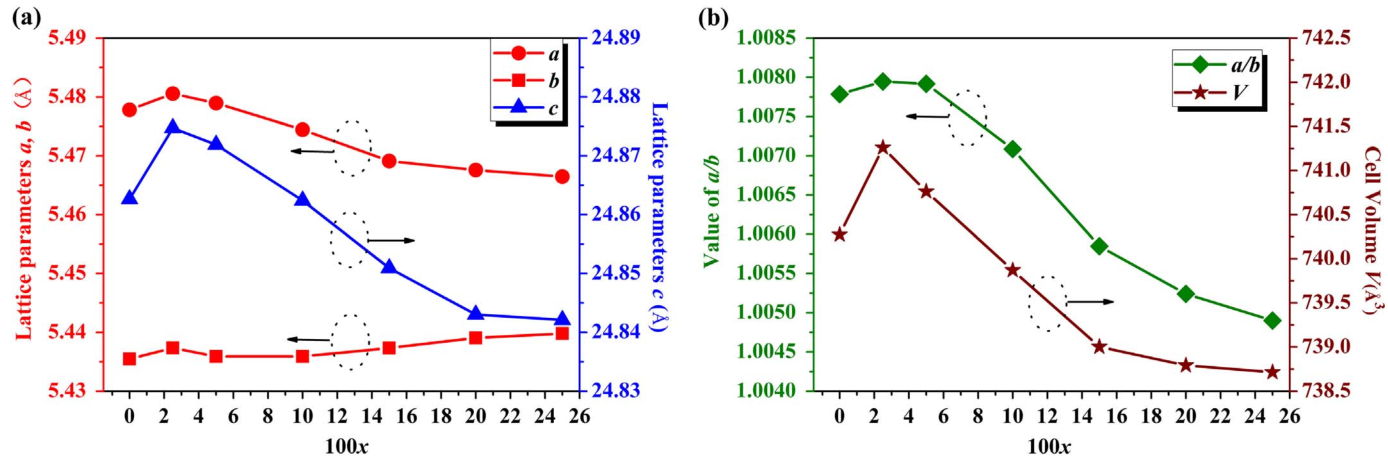

Fig. 3. Doping contents dependent (a) lattice parameters and (b) value of a/b and volume of the CBNLCN-x ceramics.

Table 1
Atomic coordinates, atomic occupancies, and lattice parameters of CBN at room temperature.

Atom	Site	X	Y	z	Occupancy
Ca1	4a	0.2555(2)	0.7628(2)	0.5	0.892
Bi1	4a	0.2555(2)	0.7628(2)	0.5	0.108
Ca2	8b	0.25	0.73339(4)	0.69996(1)	0.054
Bi2	8b	0.25	0.73339(4)	0.69996(1)	0.946
Nb1	8b	0.28606(9)	0.24280(11)	0.58377(1)	1
O1	4a	0.3361(7)	0.1902(8)	0.5	1
O2	8b	0.2678(10)	0.2700(6)	0.66582(5)	1
O3	8b	0.4878(14)	0.4742(13)	0.25862(11)	1
O4	8b	0.5289(6)	0.5240(5)	0.56754(5)	1
O5	8b	0.6287(4)	0.0555(4)	0.58250(5)	1

Lattice parameters: a = 5.47780(21) Å, b = 5.43549(21) Å, c = 24.8627(9) Å. V = 740.273(5) $Å^3$.

Table 2
Atomic coordinates, atomic occupancies, and lattice parameters of CBNLCN-5 at room temperature.

Atom	Site	X	Y	Z	Occupancy
Ca1	4a	0.2505(2)	0.7565(2)	0.5	0.836
Bi1	4a	0.2505(2)	0.7565(2)	0.5	0.114
Li1	4a	0.2505(2)	0.7565(2)	0.5	0.026
Ce1	4a	0.2505(2)	0.7565(2)	0.5	0.012
Nd1	4a	0.2505(2)	0.7565(2)	0.5	0.012
Ca2	8b	0.25	0.73120(3)	0.69987(2)	0.057
Bi2	8b	0.25	0.73120(3)	0.69987(2)	0.943
Nb1	8b	0.28410(6)	0.24533(7)	0.58407(4)	1
O1	4a	0.3701(5)	0.2295(7)	0.5	1
O2	8b	0.2441(9)	0.3178(4)	0.65709(4)	1
O3	8b	0.4840(7)	0.4815(8)	0.25265(7)	1
O4	8b	0.5333(5)	0.5442(4)	0.56309(4)	1
O5	8b	0.6019(4)	0.0573(5)	0.58359(4)	1

Lattice parameters: a = 5.47895(15)Å, b = 5.43591(15)Å, c = 24.8719(6)Å. V = 740.762(4)$Å^3$.

Table 3
Atomic coordinates, atomic occupancies, and lattice parameters of CBNLCN-10 at room temperature.

Atom	Site	X	Y	Z	Occupancy
Ca1	4a	0.2479(2)	0.7588(2)	0.5	0.784
Bi1	4a	0.2479(2)	0.7588(2)	0.5	0.116
Li1	4a	0.2479(2)	0.7588(2)	0.5	0.05
Ce1	4a	0.2479(2)	0.7588(2)	0.5	0.025
Nd1	4a	0.2479(2)	0.7588(2)	0.5	0.025
Ca2	8b	0.25	0.73164(2)	0.69972(1)	0.058
Bi2	8b	0.25	0.73164(2)	0.69972(1)	0.942
Nb1	8b	0.28406(4)	0.24218(6)	0.58381(1)	1
O1	4a	0.3503(4)	0.1997(5)	0.5	1
O2	8b	0.2239(5)	0.3136(3)	0.65629(3)	1
O3	8b	0.4871(5)	0.4793(6)	0.25061(6)	1
O4	8b	0.5427(4)	0.5295(4)	0.56304(4)	1
O5	8b	0.6043(4)	0.0627(4)	0.58194(4)	1

Lattice parameters: a = 5.47444(13) Å, b = 5.43592(13) Å, c = 24.8624(6) Å. V = 739.871(3) $Å^3$.

Table 4
Atomic coordinates, atomic occupancies, and lattice parameters of CBNLCN-15 at room temperature.

Atom	Site	X	Y	Z	Occupancy
Ca1	4a	0.2580(2)	0.7567(2)	0.5	0.744
Bi1	4a	0.2580(2)	0.7567(2)	0.5	0.106
Li1	4a	0.2580(2)	0.7567(2)	0.5	0.075
Ce1	4a	0.2580(2)	0.7567(2)	0.5	0.037
Nd1	4a	0.2580(2)	0.7567(2)	0.5	0.037
Ca2	8b	0.25	0.73146(3)	0.69964(1)	0.053
Bi2	8b	0.25	0.73146(3)	0.69964(1)	0.947
Nb1	8b	0.28284(6)	0.24343(8)	0.58355(1)	1
O1	4a	0.3231(5)	0.1819(5)	0.5	1
O2	8b	0.2759(7)	0.2870(4)	0.66115(4)	1
O3	8b	0.4854(9)	0.4906(10)	0.24944(9)	1
O4	8b	0.5327(5)	0.5363(4)	0.56644(4)	1
O5	8b	0.6026(4)	0.0692(4)	0.58261(4)	1

Lattice parameters: a = 5.46920(17) Å, b = 5.43736(16) Å, c = 24.8509(7) Å. V = 739.016(4) $Å^3$.

Bi^{3+} in the A-site decreases. The anisotropic behavior of the A-site Bi^{3+} with 6s2 lone pair electrons deforms the prototype structure along the a (b)-axis [23]. The replacement of the asymmetric Bi^{3+} by symmetric Li^+, Ce^{3+}, and Nd^{3+} reduces the orthorhombic distortion. With increasing (Li, Ce, Nd)-doping, the change in the Nb–O1–Nb bond angle and A–O1 bonds length is consistent with the variation in a/b value shown in Fig. 3.

Fig. 5 shows the SEM images of natural surfaces of CBNLCN-100x ceramics sintered at 1050 °C. The average grain sizes with error bars of CBNLCN-100x ceramics are shown in Fig. 6. The micrograph shows the presence of pore-free microstructure, indicating that dense samples are achieved under the preparation conditions and can be proofed by the inset in Fig. 6. The grain growth is found to be structurally highly anisotropic with plate-like grains in Fig. 5(a) and (b). The plate-like morphology of the grains is a characteristic feature of bismuth layer-structured compounds, because of a high grain growth rate in the direction perpendicular to the c-axis owing to the lower surface energies of the (001) planes induced predominantly in sintering [26]. However, with increasing (Li, Ce, Nd)-doping, as shown in Fig. 5(c)-(f), the grain

Table 5
Atomic coordinates, atomic occupancies, and lattice parameters of CBNLCN-20 at room temperature.

Atom	Site	X	Y	Z	Occupancy
Ca1	4a	0.2546(2)	0.7585(2)	0.5	0.708
Bi1	4a	0.2546(2)	0.7585(2)	0.5	0.092
Li1	4a	0.2546(2)	0.7585(2)	0.5	0.1
Ce1	4a	0.2546(2)	0.7585(2)	0.5	0.05
Nd1	4a	0.2546(2)	0.7585(2)	0.5	0.05
Ca2	8b	0.25	0.73339(3)	0.69952(1)	0.046
Bi2	8b	0.25	0.73339(3)	0.69952(1)	0.954
Nb1	8b	0.27663(7)	0.24196(8)	0.58361(1)	1
O1	4a	0.3114(6)	0.1926(6)	0.5	1
O2	8b	0.2697(9)	0.3005(4)	0.65789(3)	1
O3	8b	0.4823(8)	0.4846(9)	0.25555(7)	1
O4	8b	0.5380(5)	0.5325(4)	0.56704(4)	1
O5	8b	0.6010(4)	0.0717(5)	0.58253(5)	1

Lattice parameters: a = 5.46758(16) Å, b = 5.43905(16) Å, c = 24.8430(7) Å. V = 738.792(4) Å^3.

Table 6
Atomic coordinates, atomic occupancies, and lattice parameters of CBNLCN-25 at room temperature.

Atom	Site	X	Y	Z	Occupancy
Ca1	4a	0.2628(3)	0.7585(2)	0.5	0.652
Bi1	4a	0.2628(3)	0.7585(2)	0.5	0.098
Li1	4a	0.2628(3)	0.7585(2)	0.5	0.124
Ce1	4a	0.2628(3)	0.7585(2)	0.5	0.063
Nd1	4a	0.2628(3)	0.7585(2)	0.5	0.063
Ca2	8b	0.25	0.73212(3)	0.68964(9)	0.049
Bi2	8b	0.25	0.73212(3)	0.68964(9)	0.951
Nb1	8b	0.27583(9)	0.24457(9)	0.58363(7)	1
O1	4a	0.3051(12)	0.2108(10)	0.5	1
O2	8b	0.2865(7)	0.2922(5)	0.65971(4)	1
O3	8b	0.4791(7)	0.5124(7)	0.26030(4)	1
O4	8b	0.5218(11)	0.5037(9)	0.57227(7)	1
O5	8b	0.5954(7)	0.0640(8)	0.57869(9)	1

Lattice parameters: a = 5.46647(22) Å, b = 5.43979(22) Å, c = 24.8421(9) Å. V = 738.716(5) Å^3.

sizes of CBNLCN-10, 15, 20, 25 ceramics significantly reduce compared to those of CBNLCN-2.5 and CBNLCN-5 ceramics. The grain size of CBNLCN-100x ceramics in the thickness scale changes slightly and that in the length scale changes dramatically. This indicates that the excess (Li, Ce, Nd)-doping delays the diffusion mechanism during the sintering process and restrains the grain growth of CBNLCN-10, 15, 20, 25 ceramics in the direction perpendicular to the c-axis [27].

Temperature dependence of (a) dielectric permittivity and (b) loss of CBNLCN-100x ceramics at 100 kHz is shown in Fig. 7. In this study, the T_c of pure CBN is ~ 933 °C, which is same with that of pure CBN (~ 930 °C) reported by Yan et al. [6] When the x concentration is < 0.025, the T_c value slightly increases to ~ 941 °C. With the x value increasing to 0.05, the T_c value mildly decreases to ~ 938 °C. However, when the x concentration is > 0.05, there are no obvious dielectric peaks due to the sharp decrease in the resistivity and more involved flaw in (Li, Ce, Nd) co-doping CBN ceramics under a high measurement temperature [28]. The T_c value of CBNLCN-10, 15, 20, 25 ceramics continues to decrease, obtained by the valley of dielectric loss shown in Fig. 7(b). In addition, the modified CBNLCN-100x ceramics have a lower dielectric loss in the temperature range 500–600 °C (the inset of Fig. 7b), which would be significant for the high-temperature piezoelectric applications. The ferroelectric to paraelectric phase transition temperature (T_c) is closely related to their orthorhombic distortion, which is explained in terms of a lattice mismatch between BO2 and AO planes in the perovskite-type unit of AB_2O_7 [29] and the distincter the orthorhombic distortion is, the higher the T_c will be. The change in the T_c value is also consistent with the variation of the a/b value shown in Fig. 3.

Fig. 8 shows the resistivity ρ as a function of temperature and can be fitted by the Arrhenius Eq. (1):

$$\rho = A\exp(E_a/kT) \quad (1)$$

where A is the pre-exponential factor, E_a is the activation energy of charge carriers, k is the Boltzmann constant, and T is the absolute temperature. The dots are resistivity and lines are the linear fitting of the experimental data according to the Arrhenius relationship. As the concentration of (Li, Ce, Nd) increases from 0 to 2.5 mol%, the DC resistivity increases. Further increasing the concentration of (Li, Ce, Nd) decreases the DC resistivity.

The activation energy E_a at low temperature and high temperature obtained from the Arrhenius plots and the resistivity at 500 °C for CBNLCN-100x ceramics are shown in Table 8. The E_a values at low temperature are found to be in the range 0.91–1.05 eV for CBNLCN-2.5, 5, 10 and 15 ceramics, indicating that the electrical conduction process predominates by the extrinsic defects, namely, oxygen vacancies. However, when the temperature increases to 450–700 °C, the activation

Table 7
Selected bond lengths (Å) and bond angles (°) for CBNLCN-100x.

Bond type	CBN	CBNLCN-5	CBNLCN-10	CBNLCN-15	CBNLCN-20	CBNLCN-25
Nb—O						
Nb—O1—Nb (°)	158.47	154.19	156.69	158.11	162.00	166.62
Nb—O1	2.120	2.145	2.127	2.114	2.103	2.092
Nb—O2	2.047	1.872	1.874	1.945	1.873	1.909
Nb—O4	1.938	1.863	1.885	1.880	1.838	1.968
Nb—O4	2.066	2.185	2.170	2.126	2.170	1.971
Nb—O5	1.837	1.924	1.928	1.965	1.958	1.951
Nb—O5	2.135	2.019	2.007	1.990	2.001	2.008
Ca1/Bi1/Li/Ce/Nd—O (A—O)						
A—O1	3.191	3.396	3.305	3.109	3.056	3.969
A—O1	2.312	2.086	2.189	2.402	2.438	2.508
A—O1	2.365	2.654	2.461	2.339	2.382	2.471
A—O1	3.143	2.939	3.091	3.146	3.093	2.988
A—O4	2.604 × 2	2.489 × 2	2.484 × 2	2.542 × 2	2.585 × 2	2.645 × 2
A—O4	2.605 × 2	2.559 × 2	2.572 × 2	2.592 × 2	2.586 × 2	2.673 × 2
A—O5	2.305 × 2	2.452 × 2	2.389 × 2	2.415 × 2	2.401 × 2	2.365 × 2
A—O5	2.380 × 2	3.271 × 2	3.268 × 2	3.264 × 2	3.270 × 2	3.145 × 2
Bi2/Ca2—O						
Bi2/Ca2—O2	2.660	2.486	2.520	2.602	2.574	2.599
Bi2/Ca2—O2	2.775	2.920	2.821	2.765	2.828	2.725
Bi2/Ca2—O2	2.961	2.981	3.086	3.032	3.029	3.099
Bi2/Ca2—O2	3.039	3.362	3.345	3.171	3.255	3.211

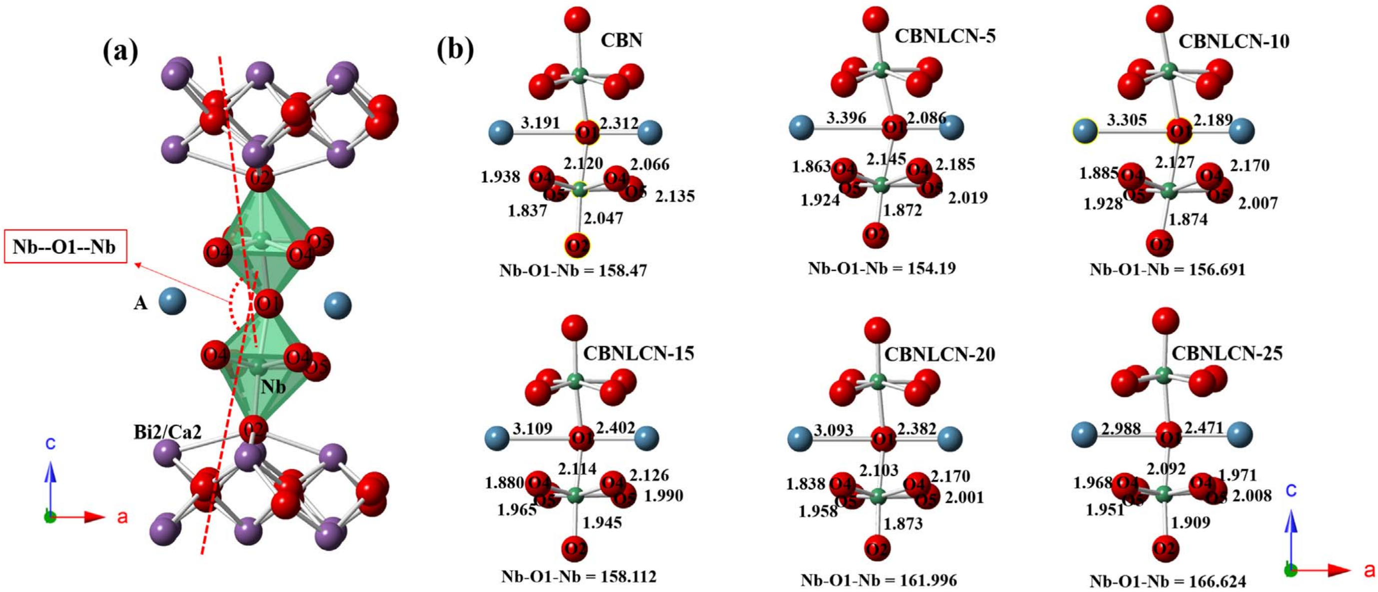

Fig. 4. The visualized crystal structure of (a) CBN and (b) the perovskite-layer units $(AB_2O_7)^{2-}$ of the CBNLCN-100x (x = 0, 0.05, 0.1, 0.15, 0.2 and 0.25) ceramics.

energy (E_a) adds to 1.26–1.86 eV, corresponding to the intrinsic conduction. The similar phenomenon is also reported in literature [30,31]. It is interesting that a significant increase in resistivity at 500 °C is observed for CBNLCN-2.5, whereas the resistivity at 500 °C of CBNLCN-5, 10, 15, 20, 25 decreases, which is also consistent with the variation in the a/b value shown in Fig. 3. The rotation and tilting of NbO_6 octahedral leads to a low mobility of defects and charge carrier [32], resulting in increase of the value of DC resistivity.

Fig. 9 shows the piezoelectric coefficient d_{33} value as a function of x contents. The d_{33} value for CBNLCN-2.5 ceramics is found to be ~ 5 pC/N, and the d_{33} values gradually increase with increasing (Li, Ce, Nd) concentration. The d_{33} value for CBNLCN-15 ceramics is found to be 13.1 pC/N, twice larger than that of CBNLCN-2.5 ceramic, although the DC resistivity of CBNLCN-15 ceramic is smaller than that of CBNLCN-

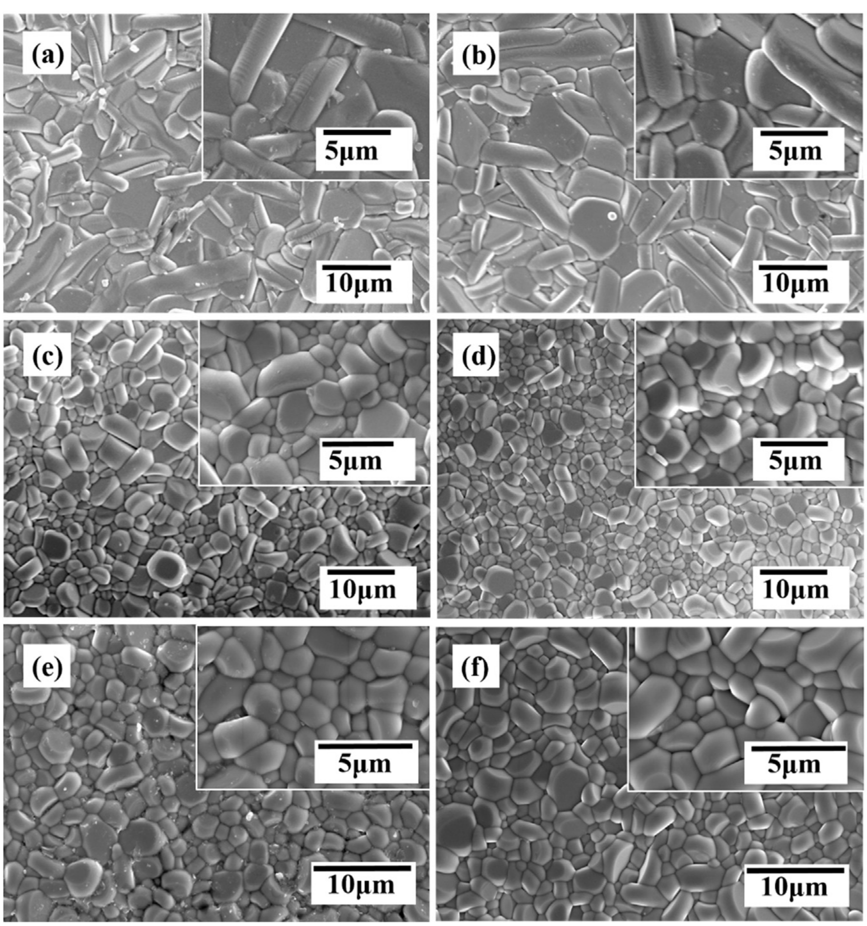

Fig. 5. The SEM images for (a) CBNLCN-2.5, (b) CBNLCN-5, (c) CBNLCN-10, (d) CBNLCN-15, (e) CBNLCN-20 and (f) CBNLCN-25 ceramics. The inset is the magnified SEM images.

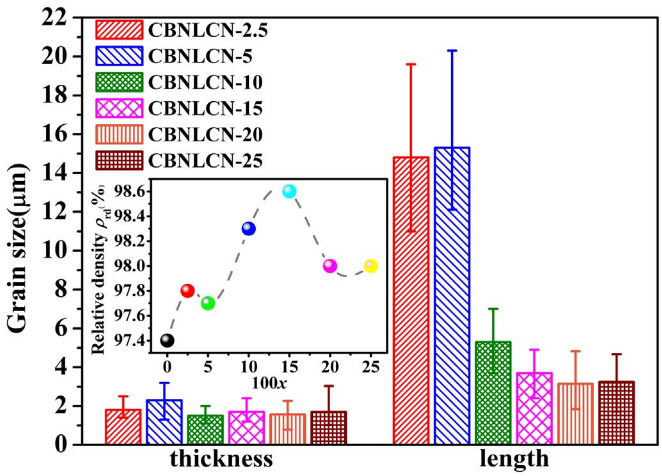

Fig. 6. Grain sizes of CBNLCN-100x ceramics. The inset shows the relative density of CBNLCN-100x ceramics.

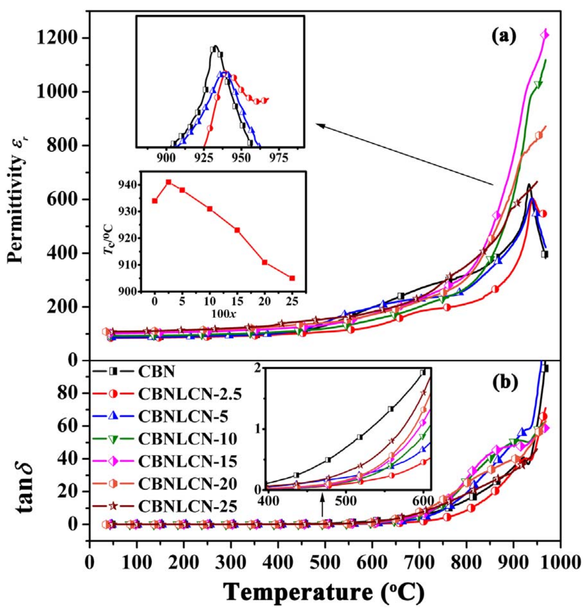

Fig. 7. Temperature dependence of (a) dielectric constant ε_r and (b) dielectric loss tanδ of CBNLCN-100x ceramics at 100 kHz.

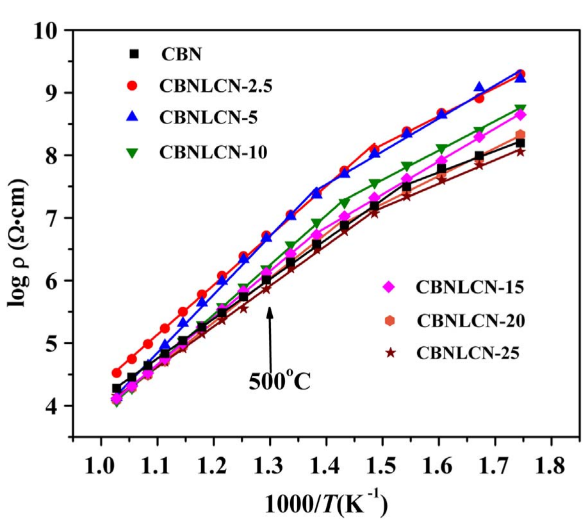

Fig. 8. Temperature dependence of DC resistivity ρ of CBNLCN-100x ceramics.

Table 8
Activation energy E_a at low temperature and high temperature (abbreviated as E_a LT and E_a HT), temperature T_0 of E_a translation obtained from the Arrhenius plots of CBNLCN-100x and the resistivity at 500 °C for CBNLCN-100x ceramics.

Compositions	E_a(eV) (LT)	T_0	E_a(eV) (HT)	ρ (Ω cm) at 500 °C
CBN	0.68	375	1.26	1.0×10^6
CBNLCN-2.5	0.91	400	1.58	5.2×10^6
CBNLCN-5	1.05	450	1.86	4.7×10^6
CBNLCN-10	1.04	425	1.59	1.5×10^6
CBNLCN – 15	1.05	450	1.50	1.3×10^6
CBNLCN-20	0.89	425	1.42	1.1×10^6
CBNLCN-25	0.76	400	1.29	7.3×10^5

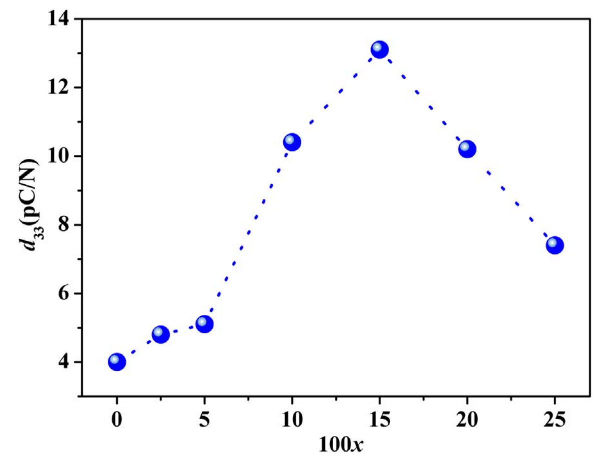

Fig. 9. Piezoelectric coefficient d_{33} values at room temperature of CBNLCN-100x ceramics.

2.5 ceramic as shown in Fig. 8. Lower DC resistivity is well known to result in a smaller polarization voltage and lower d_{33} value. The similar abnormal phenomenon has been reported in Li/Ce and Li/Y multi-modified $Na_{0.5}Bi_{2.5}Nb_2O_9$ ceramics [19]. Tian et al. [9] reported that pure CBN ceramics with small grain size have better piezoelectric properties, attributed to the decrease in the electric domain width and increase in the domain wall per unit volume, leading to more reversible domains. Decreasing grain size of CBNLCN-15 ceramics may enhance piezoelectric properties. In addition, the morphotropic phase boundary (MPB) of tetragonal and orthorhombic phase near by $100x = 15$ has a significant improvement in piezoelectric properties. The d_{33} value for CBNLCN-20, 25 ceramics gradually decrease to ~ 7.4 pC/N, attributed to the decrease of DC resistivity.

Fig. 10 shows the variation in d_{33} of (Li, Ce, Nd) multi-doped CBN ceramics with thermal depolarization temperature. For CBNLCN-2.5 and CBNLCN-5 ceramics, it is obvious that d_{33} is independent of annealing temperatures (< 900 °C). However, the d_{33} values of CBNLCN-10 and CBNLCN-15 ceramics decrease slightly as annealing temperature increases from 30 °C to 700 °C. What is more, the d_{33} values of CBNLCN-20 and CBNLCN-25 ceramics decrease sharply annealing at 400 °C. For CBNLCN-10 and CBNLCN-15 ceramics, the internal stresses increase with decreasing grain size, providing an additional driving force leading to the obvious d_{33} degradation against temperature [10]. For CBNLCN-20 and CBNLCN-25 ceramics, there would be too many defects which have a negative impact on thermal stability. For all the ceramics, the d_{33} values decrease dramatically when the annealing temperature is near T_c and become almost zero when the temperature increases above T_c. Even annealing at 850 °C for 2 h, the d_{33} value of CBNLCN-15 ceramics retains 11.3 pC/N (86% of its initial values), demonstrating that the CBNLCN-15 ceramics are a promising candidate for ultrahigh temperature applications.

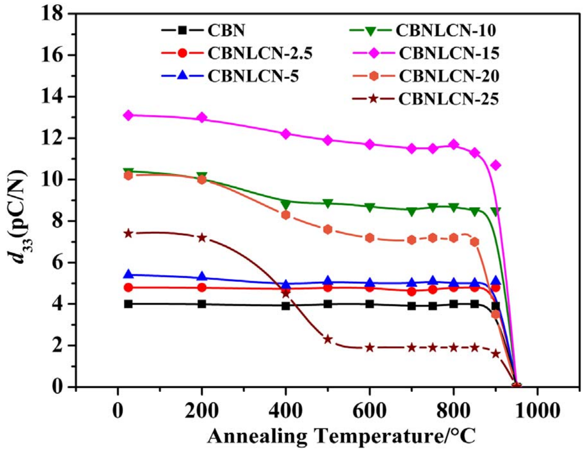

Fig. 10. Annealing temperature dependence of the piezoelectric coefficient (d_{33}) of CBNLCN-100x ceramics, at elevated temperatures with a soaking time of 2 h.

4. Conclusions

In summary, Aurivillius phase structural $CaBi_2Nb_2O_9$ ceramics with (Li, Ce, Nd) multi-dopants were successfully prepared by the conventional solid-state sintering method. XRD patterns of all compounds exhibited single phase structure, and no secondary phase was detected. There should be two types of physical mechanism in the variation of a/b. At the beginning, (Li, Ce, Nd) multi-dopants with smaller size in A site increased orthorhombic distortion. However, with increasing dopants, A-site Bi^{3+} ions were replaced by the dopants gradually, decreasing the orthorhombic distortion and leading to a tendency from orthorhombic system to tetragonal system in the crystal structure. The piezoelectric coefficient d_{33} value was significantly increased by the substitution of (Li, Ce, Nd). CBNLCN-15 ceramics possessed optimal piezoelectric properties (d_{33} ~ 13.1 pC/N, T_c > 900 °C), which would be attributed to decreasing grain sizes and morphotropic phase boundary (MPB). In addition, the piezoelectric properties of CBNLCN-15 ceramics exhibit excellent thermal stability that remained 86% of their initial d_{33} values (13.1 pC/N) after annealing at 850 °C for 2 h.

Acknowledgements

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